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Agentic AI

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Proposal: Fuel Consumption Agent

Problem Statement & Motivations

The cost of fuel has increased over the years, and students lack the resources to track the amount of gas used, to find alternative transportation methods, and better implement sustainability. This proposal would differ from traditional trackers by collecting data and actually assisting users in making actionable decisions to help save and reduce fuel use.

Target users. The primary target audience would be college commuters. Generation Z is the most environmentally aware and has proven to alter purchasing behavior, change lifestyles, and advocate for change. Additionally, this group would be ideal because of the long distances and the frequency of their journeys.

Agentic Solution. Agentic Artificial Intelligence is the future of technological advancements, and the perfect tool to develop a solution for excess fuel consumption amongst college students. An agentic solution can make autonomous decisions based on the user's input. Typically, the amount of fuel usage and the goal wished to be reached. The agent will then suggest behavioral changes or a different mode of transportation based on the user's desired goal.

Use Case & Scenarios

User One: Abby, a Student at Georgia State University, wants to reduce fuel use by 15%, but she does not know where to start. Abby would first input her average fuel use per week and her goal to reduce it by 15%. The agent will then suggest actionable decisions to reach that goal example include carpooling, driving at a steady speed, and reducing idling. At the end of the week, the agent will assess the user's feedback.

User Two: Edwin is a student at Georgia Institute of Technology, and typically drives from Sandy Springs, Georgia three times a week. He is looking for alternate modes of transportation to reduce his carbon footprint. The agent will recommend alternating between the train, bus, and driving. By the end of the work week the agent will compare the original number of fuel used to the goal the user inputted.

Initial System Design

The initial list of agents includes a planning agent, a visualization agent, and a feedback agent. However, this change list is subject to change. There might be more agents and features that could enhance user results. As for the application programming interface, I am currently planning on using Google Opal, Word, and prompt engineering using various LLMs. The data will come from user inputs asking for 1) current fuel use and 2) future goal of reduced fuel use. Google Docs, and other tools such as Google Collab. There will most likely be more additions to the APIs and data sources in phase one.

Planning agent:

- The user inputs the average weekly consumption and the desired amount of fuel they wish to use.
- The agent interprets the data input and translates it into an actionable plan.

- Ultimately, how can the user reach his/her goals?

Visualization agent:

- The visual agent will share the end of the weekly feedback with users, comparing the fuel used to the goal to be met.
- Makes information easier for students to interpret.
- Ultimately, was the goal met?

Feedback Agent:

- This agent will provide the user with feedback based on the recommendations provided and adjust itself accordingly, depending on whether the user's goal is reached.
- Checks for effective results and prevents ineffective actions.
- Ultimately, what can be improved?

Feasibility & Risk

The project could be explored by taking a more analytical and agentic approach. Given the timeline, that would not be plausible. It's also centered around students accurately implementing their information, and if not implemented accurately, it could cause hallucinations. Users must also have trust in the results to achieve their goals. Given the uncertainty around the topic of Artificial Intelligence and bias users may feel uneasy.

Timeline

- Week 1-6: Initial steps, developing front end, and multiagent workflows(milestones 1-2)
- Week 6-10: Testing and Implementation(milestone 3-final presentations)

